

SOFIYSK, Russian Federation

WMO: 314780

Lat: 52.2573N

Lon: 133.9960E

Elev: 902

StdP: 90.94

Time Zone: 10.00 (E10)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.8	-41.2	-45.6	0.0	-42.5	-43.9	0.1	-41.0	6.1	-13.1	4.4	-17.3	1.0	70	0.701

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	26.0	16.4	24.2	16.0	22.3	15.3	18.3	22.6	17.5	21.7	16.6	20.6	2.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	13.5	19.8	16.1	12.8	19.0	15.1	12.0	18.4	55.9	22.1	53.1	21.9	50.3	20.7	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	5.5	4.4	-44.9	29.2	2.0	1.4	-46.3	30.2	-47.4	31.1	-48.5	31.9	-49.9	32.9		
(5)			-44.6	19.8	1.4	0.8	-45.6	20.4	-46.4	20.8	-47.1	21.3	-48.2	21.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.5	-28.3	-24.6	-15.4	-4.2	6.2	12.6	15.9	14.0	6.8	-4.3	-18.0	-28.3
	DBStd	16.91	5.12	5.31	5.86	5.35	3.94	3.34	2.68	3.23	4.10	5.46	7.87	6.78
	HDD10.0	6099	1189	968	787	427	130	12	1	7	109	442	840	1188
	HDD18.3	8724	1447	1201	1045	677	377	174	83	138	345	701	1090	1446
	CDD10.0	428	0	0	0	0	11	90	182	129	15	0	0	0
	CDD18.3	11	0	0	0	0	0	2	6	3	0	0	0	0
	CDH23.3	241	0	0	0	0	6	62	125	48	1	0	0	0
	CDH26.7	30	0	0	0	0	1	6	21	3	0	0	0	0
Wind	WSAvg	1.7	1.1	1.5	1.8	2.0	2.4	1.8	1.7	1.5	1.8	1.8	1.4	1.1
Precipitation	PrecAvg	739	7	5	11	37	73	107	162	156	100	42	23	10
	PrecMax	974	26	13	22	74	131	194	302	351	234	93	51	28
	PrecMin	517	1	1	0	10	27	38	51	42	27	13	10	1
	PrecStd	111	5	3	6	16	28	42	62	69	41	19	9	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.4	-6.3	3.1	15.7	24.1	27.3	29.1	26.7	21.5	13.1	0.5	-6.4
		MCWB	-12.4	-7.8	-1.3	7.5	12.3	17.1	17.5	17.4	13.7	7.1	-0.9	-7.4
	2%	DB	-14.1	-9.6	-0.2	10.7	20.1	25.2	26.5	24.7	18.8	8.4	-2.2	-9.5
		MCWB	-14.9	-11.0	-3.6	4.6	10.1	15.7	16.8	16.6	12.5	4.3	-3.5	-10.4
	5%	DB	-16.9	-12.6	-2.4	7.2	17.6	22.9	24.4	22.7	16.4	5.6	-3.8	-13.0
		MCWB	-17.5	-13.7	-5.1	2.7	8.8	14.0	16.7	15.9	11.3	2.2	-4.8	-13.7
	10%	DB	-19.6	-15.0	-4.5	4.5	15.0	20.7	22.4	21.0	14.4	3.2	-5.7	-17.0
		MCWB	-20.1	-15.9	-6.7	0.4	7.8	13.5	16.3	15.7	10.5	0.6	-6.7	-17.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.4	-7.5	-0.9	8.2	13.0	19.3	19.4	19.3	15.2	7.5	-0.6	-7.5
		MCDB	-11.6	-6.5	2.8	14.7	23.0	23.7	23.6	23.6	19.3	11.2	0.7	-6.8
	2%	WB	-15.0	-11.0	-3.3	5.1	11.2	17.0	18.4	18.1	13.6	4.9	-3.2	-10.4
		MCDB	-14.3	-9.5	-0.7	10.5	18.4	23.0	22.7	22.1	17.4	8.1	-2.3	-9.4
	5%	WB	-17.5	-13.7	-5.0	2.6	9.7	15.6	17.7	17.3	12.2	2.6	-4.8	-13.7
		MCDB	-16.9	-12.7	-2.5	6.7	15.5	20.4	22.0	20.6	15.2	5.2	-3.9	-12.9
	10%	WB	-20.0	-15.9	-6.7	0.5	8.4	14.4	17.0	16.2	11.0	0.7	-6.8	-17.4
		MCDB	-19.6	-15.0	-4.5	4.0	13.8	19.1	21.1	19.5	13.9	2.6	-5.8	-16.9
Mean Daily Temperature Range	5% DB	MDBR	14.2	16.1	16.0	12.7	11.9	13.0	11.4	11.7	11.4	10.3	12.0	12.7
		MCDBR	14.2	16.1	16.1	15.4	18.3	18.1	16.5	15.9	15.6	13.9	10.2	10.4
		MCWBR	13.5	14.6	12.9	9.8	9.4	8.9	8.0	8.5	9.4	9.4	9.0	9.9
	5% WB	MCDBR	14.1	15.9	15.5	14.4	15.6	14.3	11.9	11.5	12.8	12.5	9.8	10.5
		MCWBR	13.4	14.4	12.5	9.3	8.5	7.8	6.4	6.7	8.3	8.7	8.8	10.0
Clear-Sky Solar Irradiance	taub	0.234	0.266	0.326	0.435	0.488	0.448	0.440	0.379	0.345	0.341	0.275	0.231	
	taud	2.179	2.113	1.961	1.761	1.875	2.119	2.196	2.377	2.404	2.159	2.083	2.125	
	Ebn at Noon	778	845	846	784	767	808	809	847	837	734	683	700	
	Edh at Noon	67	98	142	199	187	148	135	107	92	94	74	59	
All-Sky Solar Radiation	RadAvg	1.44	2.61	4.24	5.22	4.92	5.35	4.92	4.24	3.21	2.03	1.42	1.10	
	RadStd	0.04	0.07	0.17	0.40	0.53	0.57	0.41	0.41	0.33	0.17	0.10	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page